Chenxi Feng

Tel: +1 (607) 227-0681; Email: cf545@cornell.edu; Website: https://fcxee.github.io/

EDUCATION

Cornell University — Ithaca, NY, U.S.
Ph.D. in Atmospheric Science — *09/2024 - Present*
Thesis: Resilience and Vulnerability of Terrestrial Carbon Sequestration Under Warming Climate

Nanjing University — Nanjing, China
B.S. in Atmospheric Science — *09/2020 - 06/2024*
Thesis: Improvements and Applications of Methane Retrieval Approach Using Imaging Spectrometer

RESEARCH INTERESTS

- Climatological sensitivity of terrestrial carbon sequestration
- Climate mitigation via solar geoengineering and carbon dioxide removal
- Earth system modeling
- AI-guided observation-model fusion

PUBLICATIONS

(*, co-first author)

2026 **Feng, C.**, Zhang, S., Qin, S., Xu, X., Tao, F., Luo, Z., ... & Luo, Y. Global convergence in temperature response of microbial respiratory soil organic carbon loss. *In prep.*

2026 Li, F.*, **Feng, C.***, Juan, J. R., Guanter, L., Chen, H., Sun, S., ... & Zhang, Y. An optimized methane retrieval approach based on morphological fusion for mapping methane point emissions from spaceborne imaging spectroscopy. *Remote Sensing of Environment*. 10.1016/j.rse.2026.115718

2026 Gang, C., Wei, N., **Feng, C.**, Xu, H., Liu, H., Tao, F., ... & Luo, Y. Net primary productivity orchestrates uncertainty sources driving global soil organic carbon under land use change. *Communications Earth & Environment*. 10.1038/s43247-026-03312-6

2026 Li, F., Lin, K., Yan, Y., Bai, S., Huang, Q., **Feng, C.**, ... & Zhang, Y. Comparing the performance of different hyperspectral satellite imaging spectroscopy in mapping methane point-source emissions. *Remote Sensing of Environment*. 10.1016/j.rse.2025.115224

2024 **Feng, C.**, Chen, S., Zeng, Z.-C., Luo, Y., Natraj, V., & Yung, Y. L. Aerosol-Calibrated matched filter method for retrievals of methane point source emissions over the Los Angeles Basin. *Earth and Space Science*. 10.1029/2024EA003519

2024 Li, F., Bai, S., Lin, K., **Feng, C.**, Sun, S., Zhao, S., ... & Zhang, Y. Satellite-based surveys reveal substantial methane point-source emissions in major oil & gas basins of North America during 2022-2023. *Journal of Geophysical Research: Atmospheres*. 10.1029/2024JD040870

PRESENTATIONS

2026 **Feng, C.**, Zhang, S., Qin, S., Xu, X., Tao, F., Luo, Z., Yang, Y., Luo, Y. Substrate-induced attenuation of thermal accelerating in soil organic carbon mineralization, 2026 Ecological Society of America Meeting, 26-31 July, 2026, Salt Lake City, UT, Poster

2023 **Feng, C.**, Chen, S., Zeng, Z.-C., Luo, Y., Natraj, V., & Yung, Y. L. Assessing aerosol-induced bias in methane point source emission retrievals in the LA basin, 2023 American Geophysical Union Fall Meeting, 11-15 December, 2023, San Francisco, CA, Poster.

RESEARCH EXPERIENCES

Cornell University — Ithaca, NY, U.S.
PhD student (advisor: Dr. Daniele Visioni) — *09/2026 - Present*
Efficacy of solar geoengineering in mitigating compound hot-dry risk on terrestrial carbon sequestration

- Conducted state-of-art Earth system modeling on terrestrial carbon sequestration response to stratospheric aerosol injection
- Validated climate model simulations with observations from global dimming era (1960-1990)

Cornell University Ithaca, NY, U.S.
PhD student (advisor: Dr. Yiqi Luo) *09/2024 – 09/2026*
Global convergence in responses of soil organic carbon loss to warming

- Synthesized first global up-to-date soil incubation dataset and performed Bayesian inversion analysis
- Constrained a Community Land Model (CLM5)-type soil organic carbon (SOC) cycle model with synthesized observations
- Identified convergent underlying kinetics controlling divergent responses of microbial respiratory SOC loss to warming

California Institute of Technology Pasadena, CA, U.S.
Undergraduate Research Intern (advisor: Dr. Yuk L. Yung) *06/2023 - 01/2024*
Assessing and eliminating impacts of aerosol on methane point source emission retrieval

- Developed a fast aerosol calibration algorithm for hyperspectral remote sensing-based methane retrieval
- Applied the proposed algorithm to methane point sources in Los Angeles using remote sensing data

Nanjing University Nanjing, China
Undergraduate Research Assistant (advisor: Dr. Yongguang Zhang) *02/2022 - 07/2024*
Atmospheric methane inversion and point source quantification using hyperspectral observation

- Developed an optimized methane retrieval algorithm based on morphological fusion for spaceborne spectroscopy
- Applied the proposed algorithm to multiple satellites and detecting global energy-sector methane leakages

Nanjing University Nanjing, China
Undergraduate Research Assistant (advisor: Dr. Huilin Chen) *12/2022 - 12/2023*
In-situ detection of atmospheric methane and carbon dioxide using unmanned aerial vehicle

- Designed an unmanned aircraft methane and carbon dioxide monitoring system using gas detection sensors
- Planned and prepared in-field detection of methane emissions from wastewater treatment plants in Nanjing City

TEACHING EXPERIENCES

VIEN 1104 Introduction to Wines and Vines
Teaching Assistant, 2026 Spring semester, Cornell University (lecturer: Kathy Arnink & Ria D'Aversa)

PLSCI 3210/5210 Soil and Crop Management for Sustainability
Teaching Assistant, 2025 Fall semester, Cornell University (lecturer: Dr. Harold Van Es)

PLSCI 2600 Soil Science
Teaching Assistant, 2025 Fall semester, Cornell University (lecturer: Dr. Jonathan Russell-Anelli)

PLSCI 6250 Data-Model Fusion in Research
Teaching Assistant (voluntary), 2025 Fall semester, Cornell University (lecturer: Dr. Yiqi Luo)

ACHIEVEMENTS AND AWARDS

2024	Outstanding Thesis Award (The First Prize), Nanjing University
2024	Chenxue Scholarship for Overseas Study, Nanjing University
2023	Scholarship of Academic Excellence (The First Prize, Top 5%), Nanjing University
2022	Zhenggang Fellowship for Oversea Studies of Nanjing University (Top 5%), Nanjing University
2022	Scholarship of Academic Excellence (The Second Prize, Top 10%), Nanjing University
2022	Outstanding Student for Theory and Methods of Land Surface Remote Sensing Inversion Summer School 2022, Beijing Normal University
2021	Scholarship of Academic Excellence (The Second Prize, Top 10%), Nanjing University